Chapter 2

Lemongrass: a traditional ethno-medicinal plant of India

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Abstract: Lemongrass, a medicinal plant belonging to the family Poaceae. This medicinal plant contains compounds that can control pathogens and enhance the resistance of herbal remedies against diseases caused by pathogens. Lemongrass is commonly utilized in the preparation of herbal teas, non-alcoholic beverages, baked goods, and confections. Additionally, the essential oil derived from lemongrass is widely used as a fragrance in perfumes and cosmetics, including creams and soaps. The essential oil of lemongrass is rich in citral, which is a valuable source to produce beta carotene and vitamin A, among other substances. It contains considerable group of flavonoids, essential oils, phenolic compound, and other photochemical constituents which possesses pharmacological activities such as anti-obesity, antibacterial, anti-fungal, anti-nociceptive, anti-oxidant, anti-diarrheal, and anti-inflammatory properties which could enhance health. In addition, lemongrass oil also contains essential minerals such as Fe, Zn, Mg, Na, K,Ca, Mn, and P but unfortunately, it is not utilized in broad spectrum among rural & urban people. Keeping the importance of Lemongrass, an attempt has been made to gather the information on it through a literature survey. The results are presented in this chapter to bring attention towards its importance.

Keywords: Economic values, lemon grass, nutraceutical values

Introduction: Lemon grass, popularly known as citronella grass, is a plant species that belongs to the family Poaceae and to the genus *Cymbopogon*. The genus *Cymbopogon* is composed of about 140 species. Lemon grass is native to the semi-arid and tropical areas of Asia, America, and Africa. It is native to Pakistan, India, and Sri Lanka. In India it is in Western Ghats (Kerala, Maharashtra), Karnataka and the Tamil Nadu states other than foot slopes of the Sikkim and Arunachal Pradesh (Ranade and Thiagarajan 2015; Wifek et al. 2016). Several species of lemon grass such as *Cymbopogon bombycinus, Cymbopogon ambiguus, Cymbopogon obtectus, Cymbopogon refractus, Cymbopogon citratus, Cymbopogon citrate, Cymbopogon nardus, Cymbopogon schoenanthus* etc., found in

countries such as Australia, China, India, Africa, and others. These are perennial aromatic grass with a distinctive lemon-like odour. The name 'lemon' is derived from the citral monoterpenes it contains. The members of the *Cymbopogon* genus produce volatile oils and thus are also known as aromatic grass (Sharma et al. 2019). The *Cymbopogon* essential oils are characterized by the monoterpene constituents like limonene, citral, elemol, citronellal, 1,8 cineole, citronellol, linalool, geraniol, methylheptenone, b-carophyllene, geranylformate and geranyl acetic acid derivation. Lemongrass is a coarse plant with tufted linear leaves that grows in thick bunches from a strong base, reaching a height of about 3 meters with a width of about 1 meter. In addition to its culinary usage, lemongrass offers a wide array of medicinal benefits and is in high demand due to its antimicrobial properties (Wifek et al. 2016).

Description: It is a perennial monocotyledonous grass that can grow up to 6 feet tall and 4 feet wide (Figure 1). It grows in clumps. The culm is stout, erect, 1.8 to 2 meter high. It has long, slender, drooping, bright green leaves 1.3-2.5 inches wide and 3 feet long and 3feet (Figure 2). The leaves are simple, glaucous, linear tapering upwards and along the margins; ligule very short; sheaths terete, those of the barren shoots widened and tightly clasping at the base, others narrow and separating and have entire margins. The inflorescence is a long spike about one metre in length. The flowers grow in panicles. The inflorescence is 30-60 cm long (Figure 3). The arrangement of the flowers of this fragrant grass gives it the name '*Cymbopogon'* (Saxena and Brahmam 1996; Wifek et al. 2016).

Phytochemical constituents of Lemongrass: The analysis of Cymbopogon through qualitative and quantitative phytochemical screening has identified significant bioactive chemical compounds that may be associated with the plant's therapeutic effectiveness. Various bioactive constituents, including ketones, alcohols, phenols, terpenes, flavonoids, saponins, steroids, tannins, alkaloids, terpenoids, polyphenols, esters, aldehyde, and fatty acids, have been identified. According to research, the most crucial compounds found in *Cymbopogon* are essential oil and flavonoids, which play a key role in the plant's therapeutic and pharmacological properties (Asif and Khodadadi 2013; Roriz et al. 2014; Zannou et al. 2015). The primary source of essential oil extracted from lemongrass is predominantly located in the plant's leaves, which contain up to 5 % dry weight of essential oils. The main component, citral, gives the oil a unique lemon-like fragrance (Kaini et al. 2022).



Figure 1: Vegetative parts of Lemongrass



Figure 2: Leaves of Lemongrass



Figure 3: Inflorescence of Lemongrass

Economic importance: The leaves are the source of the essential oil, which contains over 70% citral. This citral is a key component for the synthesis of B-ionones. Additionally, the oil includes more than 17% oleoresin. Citral is widely used in perfumery for various products such as soaps, detergents, cosmetics, insect repellents, room fresheners, ayurvedic preparations, and flavouring agents for soft drinks and tea. The oil, rich in citral, possesses germicidal, medicinal, and flavouring properties. Therefore, cultivating lemon grass not only ensures a steady income for farm families year-round but also fills open spaces in plantations. Due to its ease of cultivation, low risk, minimal investment, and guaranteed market and income. In addition to the domestic market, lemongrass oil also has significant export potential. Currently, Sri Lanka and Indonesia are the top exporters of these oils (Joy et al. 2006; Sharma et al. 2022). Lemongrass is cultivated worldwide, with the main sources of essential oil being India and Guatemala. The East Indian variety from India contains a higher citral content compared to the West Indies variety from Guatemala. India, as the leading global producer of lemongrass, contributes nearly 80% of the total output, with a monthly market flow of 45 to 55 MT. Indian lemongrass is typically harvested three times a year, but some well-maintained crops can yield up to 5 harvests annually. It is important to note that leasing land accounts for approximately 25-30% of the total production costs in India. This is significant due to the increasing population in India, which has led to the urbanization and relocation of many lemongrass farming areas (https://www.berjeinc.com/media/berje-report/lemongrass-production-in-india/).

Traditional importance: In the continents of Asia, South America, and Africa, the leaves have traditionally been utilized as tea or decoction. These leaves contain essential bioactive compounds that determine the plant's various ethno-medicinal properties. Additionally, they are employed as deodorants in numerous products such as perfume, local soaps, candles, and other insect repellents. Moreover, in certain regions of Asia and African countries, they have been employed as snake and reptile repellents (Oladeji et al. 2020).

Food values: Dried lemongrass leaves are widely used as a lemon flavour ingredient in herbal teas, prepared either by decoction or infusion of 2-3 leaves in 250 or 500 ml of water and other formulations. Lemongrass tea is a diuretic and imparts no biochemical changes to the body in comparison with the ordinary tea. Lemongrass iced tea is prepared by steeping several stalks in a few quarts of boiling water. This can also be combined with green or black teas. Lemongrass is commonly used in Asian cookery. When Thai food was embraced in

the US, lemongrass became a household name. A simple syrup made by steeping lemongrass in a mix of equal parts hot water and sugar can be used to enhance fruit salads (Joy et al. 2006). A blend of lemongrass, garlic, ginger, and oil will be stable in the freezer during winter. This paste can be fried until fragrant and then cooked down with a can of coconut milk (strain to remove tough lemongrass fibres) for delicious sauce for noodle, vegetable, or seafood dishes. Lemongrass oil is used in culinary flavouring. It is used in most of the major categories of food including alcoholic and non-alcoholic beverages, frozen dairy desserts, candy baked foods, gelatines, and puddings, meat and meat products and fat and oils. It is used to improve the flavour of some fish, and can be used to flavour wines, sauces etc. Lemongrass oleoresin is mainly used in flavouring foods, drinks, and bakery preparations (Joy et al. 2006; Thasrin and Anitha 2023).

Nutraceutical potentials of Lemongrass: Lemongrass can be used either fresh or dried and powdered as a food flavouring. It is commonly added to teas, soups, and curries, and can be paired with poultry, fish, beef, and seafood. Numerous studies have highlighted the therapeutic benefits of lemongrass leaves and other parts when infused to alleviate nausea, stomach aches, constipation, prevent ulcers, and combat various stomach infections. Lemongrass essential oil is a volatile and chemically lipophilic thick yellow liquid. It primarily consists of monoterpenes, sesquiterpenes, and phenylpropanoids. The main component, citral, is composed of geranial and neral isomers and serves as a precursor to produce alpha-, beta-, and beta-carotene. Beta ionone is utilized in the synthesis of vitamin A, while alpha ionone finds applications in flavors, cosmetics, and fragrances (Fokom et al. 2019). Moreover, lemongrass essential oil exhibits potential antibacterial, anti-inflammatory, analgesic, antioxidant, antispasmodic, antipyretic, diuretic, insecticidal, larvicidal, pesticidal, antifeedant activities, and sedative properties. It is not only used as a spice for its pleasant taste but also plays a role in food preservation and processing (Boukhatem et al. 2014; Barbosa et al. 2008). The primary chemical constituents found in Cymbopogon citratus are terpenes, alcohols, ketones, aldehydes, and esters. Additionally, it contains other phytoconstituents such as Terpinolene, Geranyl acetate, Citral, Nerol Geraniol, Citronellal, Myrecene, and Terpinol Methyl heptenone, which are present in its essential oils. Furthermore, the plant has been identified to contain phenolic compounds like luteolin, isoorientin 2'-O-rhamnoside, quercetin, kaem (Thasrin and Anitha 2023).

Medicinal uses: Less reports are available on medicinal grass species (Mishra et al. 2022; Nayak and Kumar 2023). In this aspect, documentation of medicinal uses of lemongrass is an important work. Lemongrass essential oil has been utilized as medicine for centuries to enhance circulation, regulate menstrual cycles, aid digestion, and boost immunity. It is also employed in the production of perfumes, flavors, detergents, and pharmaceuticals (Majewska et al. 2019). It offers numerous benefits in preventing gastrointestinal disorders, including gastric ulcers, as well as aiding in bowel stimulation and improving digestion. Its anti-inflammatory properties are also effective in treating constipation, ulcerative colitis, diarrhoea, nausea, and stomach aches (Madane 2024). Moreover, lemongrass plays a significant role in promoting better sleep by calming the muscles and nerves, thus inducing deep sleep. It is widely utilized in Ayurvedic medicine for its healing effects on respiratory disorders such as coughs and colds. Its vitamin C content, combined with other beneficial components, provides relief from nasal blockages, flu, and bronchial asthma, among other respiratory ailments (Hanif et al. 2010; Khan et al. 2012). Lemongrass is recognized as a febrifuge and is commonly referred to as 'fever grass' due to its positive impact on reducing fevers. The antipyretic and diaphoretic properties of lemongrass are widely utilized in Ayurvedic medicine to treat fevers by promoting sweating. Lemongrass acts as an antiseptic and is effective in addressing infections like ringworm, sores, Athlete's Foot, scabies, and urinary tract infections because of its antimicrobial and anti-fungal characteristics (Coelho et al. 2016). Lemongrass aids in relieving pain and discomfort associated with headaches and migraines, enhance blood circulation, and help alleviate spasms, muscle cramps, sprains, and backaches. Lemongrass is beneficial in treating sports injuries, including dislocations, internal injuries, and bruises. It also helps in stimulation of the mind and assists in managing convulsions, nervousness, vertigo, and various neuronal disorders such as Alzheimer's and Parkinson's disease. Lemongrass is utilized in therapeutic baths to calm the nerves and alleviate symptoms of anxiety and fatigue induced by stress (Wifek et al. 2016; Madane 2024).

Conclusion: Medicinal herbs have significantly enhanced the quality of primary healthcare by providing herbal drugs that are free from any adverse health effects. The evaluation of various plants used in India's traditional therapeutic system is gaining popularity in the field of pharmacology. Lemongrass, known for its valuable essential oils and its extensive use in traditional medicine and culinary practices, is becoming increasingly popular. Given the growing interest in natural products derived from lemongrass, a comprehensive analysis of

its phytochemical and pharmacological properties is necessary. To enhance the phytochemical and pharmacological research, factors like propagation methods, extraction techniques, and harvesting times need to be carefully monitored and addressed. This will improve the physicochemical composition and biological properties of *Cymbopogon* extracts, ultimately increasing its economic value. However, further research in the form of clinical trials is required to validate the laboratory findings regarding lemongrass's potent antioxidant potential.

References

Asif A and Khodadadi E. (2013). Medicinal uses and chemistry of flavonoid contents of some common edible tropical plants. Archives of Advances in Biosciences. 4(3): 119-138.

Barbosa LCA, Pereira UA, Martinazzo AP, Maltha CRA, Teixeira RR and Melo EDC. (2008) Evaluation of the chemical composition of Brazilian commercial *Cymbopogon citratus* (D.C.) Stapf samples. Molecules. 13: 1864–1874.

Boukhatem MN, Kameli A, Ferhat MA, Saidi F and Tayebi K. (2014). The food preservative potential of essential oils, is lemongrass the answer? Journal für Verbraucherschutz und Lebensmittelsicherheit. 9: 13–21.

Coelho M, Rocha C, Cunha LM, Cardoso L, Alves L, Lima RC and Pintado M. (2016). Influence of harvesting factors on sensory attributes and phenolic and aroma compounds composition of *Cymbopogon citratus* leaves infusions, Food Research International. 89: 1029–1037.

Fokom R, Adamou S, Essono D, Ngwasiri DP, Eke P, Teugwa Mofor C, Tchoumbougnang F, Boyom Fekam F, Amvam Zollo PH, Nwaga D and Sharma AK. (2019) Growth, essential oil content, chemical composition and antioxidant properties of lemongrass as affected by harvest period and arbuscular mycorrhizal fungi in field conditions. Industrial Crops and Products. 138: 111477.

Hanif MA, Bhatti HN, Jamil MS, Anjum RS, Jamil A and Khan MM. (2010). Antibacterial and antifungal activities of essential oils extracted from medicinal plants using CO2 supercritical fluid extraction technology. Asian Journal of Chemistry. 22(10): 7787.

https://www.berjeinc.com/media/berje-report/lemongrass-production-in-india/.

Joy PP, Skaria BP, Mathew S, Mathew G, Joseph A and Sreevidya PP. (2006). Lemongrass. Indian Journal of Arecanut, Spices and Medicinal Plants. 2:55–64.

Khan MM, Iqbal M, Hanif MA, Mahmood MS, Naqvi SA, Shahid M and Jaskani MJ. (2012). Antioxidant and antipathogenic activities of citrus peel oils. Journal of Essential Oil-Bearing Plants. 15(6): 972-979.

Kiani HS, Ali A, Zahra S, Hassan ZU, Kubra KT, Azam M and Zahid HF. (2022). Phytochemical Composition and Pharmacological Potential of Lemongrass (*Cymbopogon*) and Impact on Gut Microbiota. Applied Chemistry. 2(4):229-246.

Madane SG. (2024). Review on benefits of lemon grass. World Journal of Pharmaceutical and Life Science. 10(2): 129-132.

Majewska E, Kozlowska M, Gruszczynska-Sekowska E, Kowalska D and Tarnowska K. (2019). Lemongrass (*Cymbopogon citratus*) essential oil: extraction, composition, bioactivity and uses for food preservation-a review. Polish Journal of Food and Nutrition Sciences. 69(4).

Mishra S, Agrawal B, Rathore S, Mishra AK and Kumar S. (2022). Medicinal grasses of Eastern Ghats of Odisha, India. Indian Forester. 148(11): 1172-1174.

Nayak S and Kumar S. (2023). Medicinal plants used by tribals of Odisha. State Medicinal Plants Board & Ambika Prasad Research Foundation, Odisha; APRF Publishers, India. Pp 1-672.

Oladeji OS, Adelowo FE, Ay-odele DT and Odelade KA. (2019). Phytochemistry and pharmacological activities of *Cymbopogon citratus*: A review. Scientific African. 6: e00137.

Ranade SS and Thiagarajan P. (2015). Lemon Grass. International Journal of Pharmaceutical Sciences Review and Research. 35(2): 162-167.

Roriz C, Barros L, Carvalho A, Santos-Buelga C and Ferreira C. (2014). *Pterospartum tridentatum*, *Gomphrena globose* and *Cymbopogon citratus*: a phytochemical study focused on antioxidant compounds. Food Research International. 62: 684-693.

Saxena H.O. and Bramham M. (1996). The Flora of Orissa, Vol IV, Orrisa Forest Development Corporation Ltd & Regional Research laboratory, Bhubaneswar, Odisha.

Sharma RS, Kumar Y, Bhise RN, Choudhri HPS, Verma DK, Srivastava RK and Kumar S. (2022). Economics of Lemongrass Cultivation in Jharkhand State. Economic Affairs. 67(9): 01-06.

Sharma S, Habib S, Sahu D and Gupta J. (2019). Chemical properties and therapeutic potential of citral, a monoterpene isolated from lemongrass. Medicinal Chemistry. 17:2–12.

Thasrin F and Anitha V. (2013). Nutritional and Nutraceutical Potentials of Lemongrass (*Cymbopogon citratus*). International Journal of Current Science Research and Review. 6(5): 2881-2886.

Wifek M, Saeed A, Rehman R and Nisar S. (2016). Lemongrass: A review on its botany, properties, applications, and active components. International Journal of Chemical and Biological Sciences. 9:79-84.

Zannou A, Christian TR, Gbaguidi MA and Ahoussi-Dahouenon E. (2015). Antimicrobial activity of extracts from *Cymbopogon citratus* L. and of Mentha spicata L. against fungal and bacterial strains isolated from peuhl's cheese (Waragashi) produced in Benin. International journal of Advance Research. 3: 1684-1695.